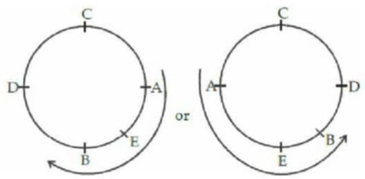
Select the correct answer using the codes given below :

(a) I only
(b) II only
(c) Both I and II
(d) Neither I nor II

Ans. (c) From the given conditions two cases are possible as shown below :



Hence, In any case, both the conditions are true.

50. Geeta is older than her cousin Meena. Meena's brother Bipin is older than Geeta. When Meena and Bipin visit Geeta, they like to play chess. Meena wins the game more often than Geeta .

Based on the above information, four conclusions, as given below, have been made. Which one these logically follow from the information given above?

(a) While playing chess with Geeta and Meena, Bipin often loses.
(b) Geeta is the oldest among the three.
(c) Geeta hates to lose the game .
(d) Meena is the youngest of the three .

Ans. (d) According to age,

Bipin > Geeta > Meena

Hence, Meena is the youngest of the three.

51. There are five hobby clubs in a college viz., photography, yachting, chess, electronics and gardening. The gardening group meets every second day, the electronics group meets every third day, the chess group meets every fourth day, the yachting group meets every fifth day and the photography group meets every sixth day. How many times do all the five groups meet on the same day within 180 days ?

(a) 3 (b) 5
(c) 10 (d) 18

Ans. (a) The L.C.M. of 2, 3, 4, 5 and 6

$= 60$

Hence,

The required number of times $= \frac{180}{60}$

$= 3$

52. A, B, C, D and E belong to five different cities P, Q, R, S and T (not necessarily in that order). Each one of them comes from a different city, Further it is given that

1. B and C do not belong to Q.
2. B and E do not belong to P and R.
3. A and C do not belong to R, S and T.
4. D and E do not belong to Q and T .

Which one of the following statements is not correct?

(a) C belongs to P
(b) D belongs to R
(c) A belongs to Q
(d) B belongs to S

Ans. (d)

	P	Q	R	S	T
A	×	✓	×	×	×
B	×	×	×	×	✓
C	✓	×	×	×	×
D	×	×	✓	×	×
E	×	×	×	✓	×

According to the chart, 'B' belongs to 'T'

Hence,

The statement "B belongs to S" is not correct.

53. Seven men, A, B, C, D, E, F and G are standing in a queue in that order. Each one is wearing a cap of a different colour like violet, indigo, blue, green, yellow orange and red. D is able to see in front of him green and blue, but not violet. E can see violet and yellow, but not red. G can see caps of all colors other than orange. If E is wearing an indigo coloured cap, then the colour of the cap worn by F is

(a) blue (b) Violet
(c) Red (d) Orange

Ans. (c) According to order:-

A → Yellow
B → Blue
C → Green
D → Violet
E → Indigo
F → Red
G → Orange

Hence,

The colour of the cap worn by F is red.

54. There are some balls of red, green and yellow colour lying on a table. There are as many red ball as there are yellow balls. there are twice as many yellow as there are green ones. The number of red balls.

(a) is equal to the sum of yellow and green balls.
(b) is double to the number of green balls.
(c) is equal to yellow balls minus green balls.
(d) cannot be ascertained.

Ans. (b) Let the number of green balls = x

$\therefore$ The number of yellow balls = 2x

$\because$ The number of red balls = The number of yellow balls.

$\therefore$ The number of red balls = 2x

Hence,

The number of red balls is double to the number of green balls.

55. In a class of 45 students, a boy is ranked 20[th]. When two boys joined, his rank was dropped by one. What is his new rank from the end ?

(a) 25[th] (b) 26[th]
(c) 27[th] (d) 28[th]

Ans. (c) Required Rank = (45 + 2) – (20 +1) + 1

= 47 – 21 + 1 = 48 – 21 = 27th

BASIC NUMERICAL ABILITY

56. In a rare coin collection, there is one gold coin for every three non-gold coins. 10 more gold coins are added to the collection and the ratio of gold coins to non-gold coins would be 1 : 2. Based on the information, the total number of coins in the collection now becomes.

(a) 90 (b) 80
(c) 60 (d) 50

Ans. (a) Let the number of gold coins = x
and the number of non-gold coins = 3x
According to question,

$\frac{x+10}{3x} = \frac{1}{2}$

$2x + 20 = 3x$
$3x + 2x = 20$
$x = 20$
The total number of coins in the collection row becomes = (x + 10) + 3x
= (20 + 10) + 3 × 20
= 30 + 60 = 90

57. A gardener has 1000 plants. He wants to plant them in such a way that he number of rows and the number of columns remains the same. What is the minimum number of plants that he needs more for the this purpose ?

(a) 14 (b) 24
(c) 32 (d) 34

Ans. (b) The number of required plants = $(32)^2$ – 1000
1024 – 1000 = 24.

58. A sum of Rs. 700 has to be used to give seven cash prizes to the students of a school for their overall academic performance. If each Prize is Rs. 20 less than its preceding prize, what is the least value of the prize ?

(a) Rs. 30 (b) Rs. 40
(c) Rs. 60 (d) Rs 80

Ans. (b) Let the least value of the prize = x ₹
According to question, x + (x + 20) + (x + 40) + (x + 60) + (x + 80) + (x + 100) + (x + 120) = 700
7x + 420 = 700 ⇒ 7x = 700 – 420
⇒ 7x = 280

$\therefore x = \frac{280}{7} = 40$

59. Out of 120 applications for a post, 70 are mal and 80 have a driver's license. What is the ratio between the minimum to maximum number of males shaving driver's license ?

(a) 1 to 2 (b) 2 to 3
(c) 3 to 7 (d) 5 to 7

Ans. (c) According to question n (A ∪ B) = n(A) + n(B) – n (A∩B)
⇒ 120 = 70 + 80 – n(A ∩ B)
⇒ n(A ∩ B) = 150 – 120 = 30
$\therefore$ Required Ratio = 30 : 70 = 3 : 7

60. In a garrison, there was food for 1000 soldiers for one month. After 10 days, 1000 more soldiers joined the garrison. How long would the soldiers be able to carry on with the remaining food ?

(a) 25 days (b) 20 days
(c) 15 days (d) 10 days

Ans. (d) According to question, $M_1D_1 = M_2D_2$
$1000 \times (30 - 10) = (1000 + 1000) \times x$

$1000 \times 20 = 2000 \times x$ $\therefore x = \frac{20,000}{2000} = 10$

61. The tank-full petrol in Arun's motor-cycle lasts for 10 days. If he starts using 25% more every day, how many days will the tank-full petrol last?

(a) 5 (b) 6
(c) 7 (d) 8

Ans. (d) Let the capacity of tank = 1000 lit
According to question,
$\because$ Petrol used in 10 day = 1000 lit

$\therefore$ Petrol used in one day $= \frac{1000}{10}$

= 100 lit

When Arun starts using 25% more every day, then petrol used in one day = 125 lit
Hence,

The required number of days $= \frac{1000}{125}$

= 8 days.

62. A person can walk a certain distance and drive back in six hours. He can also walk both ways in 10 hours. How much time will he take to drive both way ?

(a) Two hours
(b) Two and a half hours
(c) Five and a half hours
(d) Four hours